

Batch: A-4 **Roll No.:** 16010122151

Experiment / Assignment / Tutorial No: 4

TITLE: Practical Application of Wireframing Models for User Interface Design.

Objective:

Understand various types of wireframes and develop a wireframe for sample application.

Expected OUTCOME of Experiment: CO 3

Practical Application of Wireframing Models for User Interface Design

Books/ Journals/ Websites referred:

- <https://www.figma.com/>
- <https://www.figma.com/design/CNzNCPdpRPzELo6tLBFmyW/payment-process-in-e-commerce-website--Community-?t=PdZD420xdyBEdBsY-1>
- <https://miro.com/templates/low-fidelity-wireframes/>

Background Theory:

Wireframing is a fundamental step in the user interface (UI) and user experience (UX) design process. It involves creating a basic visual guide that represents the skeletal framework of a website, application, or product. Wireframes are typically low-fidelity, focusing on layout, content placement, functionality, and user interaction rather than visual aesthetics or detailed graphics. They serve as a bridge between abstract ideas and concrete designs.

Key aspects of wireframing include:

- **Purpose:** To define the structure and hierarchy of information, establish the relationships between different parts of the interface, and outline the user flow. They help in planning the layout, navigation, and content organization.
- **Fidelity Levels:**
 - **Low-fidelity wireframes:** Often hand-drawn sketches or very basic digital representations. They are quick to create and ideal for brainstorming and early conceptualization. Focus is purely on structure and content.

- **Mid-fidelity wireframes:** More detailed than low-fidelity, often created with digital tools. They include more specific elements like actual text, icons, and placeholder images, but still avoid colors and detailed styling. They are good for testing basic user flows.
- **High-fidelity wireframes (sometimes blurring into mockups/prototypes):** Closest to the final product in terms of visual detail, but still primarily focused on structure and functionality. They might include specific fonts, colors, and detailed UI elements.
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- **Benefits:**
 - **Early Feedback:** Allows stakeholders and users to review and provide feedback on the layout and functionality before significant design and development effort is invested.
 - **Cost-Effective:** Changes are much cheaper and faster to implement at the wireframing stage than later in development.
 - **Clarity:** Provides a clear visual representation of the product's structure, reducing ambiguity among team members.
 - **Focus on Functionality:** By stripping away visual distractions, wireframes force a focus on the core functionality and user flow.
- **Common Elements:** Placeholders for images, text blocks, buttons, navigation menus, input fields, and other interactive elements.
- **Tools:** Various software tools exist for wireframing, ranging from simple drawing applications to dedicated wireframing and prototyping software. Examples include Balsamiq, Figma (with wireframing kits), Axure RP, Sketch, and Adobe XD.

Understanding how to effectively create and use wireframes is crucial for computer engineers, as it enables them to communicate design requirements clearly, collaborate efficiently with designers, and contribute to building user-friendly software.

Laboratory Work:

Activity:

1. Choose the application under consideration for the wireframing project. You will design the main screens for this application.

Example scenario for reference only.

- **Scenario A: Simple E-commerce Product Page:** Design a product detail page for an online store, including product image, name, price, description, "add to cart" button, quantity selector, and related products section.
- **Scenario B: Basic To-Do List Application (Web or Mobile):** Design the main screen for a To-Do list app, allowing users to add tasks, mark them as complete, and view existing tasks. Consider basic navigation.

- Scenario C: User Profile Page for a Social Media App: Design a user's profile page, including profile picture, name, bio, number of followers/following, and tabs for different content types (e.g., posts, photos).
2. For your chosen scenario, list the **key functionalities and content elements** that must be present on the main screen(s).
 - **Sketch initial low-fidelity wireframes:** On paper or using a simple digital drawing tool, quickly sketch out 2-3 different layout ideas for your main screen(s). Focus on the placement of major elements. Choose the best layout to proceed with. A total of minimum 6 wireframes representing the entire flow of the user interaction needs to be sketched.
 - **Define User Flow (if applicable):**
 - For Scenario A, consider adding an item to the cart.
 - For Scenario B, consider the flow for adding a new task.
 - For Scenario C, consider editing the details or inviting, accepting or rejecting requests from other users
 3. Using your chosen wireframing tool, create mid-fidelity wireframes for the main screen(s) of your application.
 - Elements to include:
 - Headers, footers, navigation bars (if applicable)
 - Placeholders for images (e.g., rectangles with an "X")
 - Text blocks (e.g., "Lorem ipsum" or descriptive labels)
 - Buttons (e.g., "Add to Cart," "New Task," "Edit Profile")
 - Input fields (e.g., search bar, task input)
 - Lists or grids for displaying content
 - Icons (basic shapes or generic icons)
 4. Focus on:
 5. Layout and spacing: How elements are arranged on the screen.
 6. Information hierarchy: Which elements are most prominent?
 7. User flow: How a user would interact with the elements to complete a task.
 8. **Document each interaction tabulate meticulously:**

Name of the Product under consideration	
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Reasons for hesitation	
Reasons for mistakes	
Source for confusion	
Time required to complete the task compared while using the system / interface	☐ Less ☐ Same (no change) ☐ More
Help and assistant	☐ Easily available ☐ Intuitive (no need of assistance) ☐ Difficulty in locating the help to complete the task in hand ☐ Words used in feedback were confusing
Product worked as expected	☐ Yes ☐ No

9. Identify common themes or recurring types of usability issues across different products.
10. Reflect on how these issues could have been prevented during the design and engineering phases.

1. Application Selection and Definition

Begin by choosing a real-world application scenario for your wireframing project.

Example Selection:

- Product: Flipkart E-commerce Mobile App
 - Justification: This application serves millions of users daily and has complex flows for browsing, purchasing, and order management, making it an ideal candidate for wireframing analysis.
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2. Identify and List Key Screen Functionalities

Detail all key functions and content needed for primary screens. This allows you to scope the project precisely and ensures no mission-critical feature is overlooked.

For Scenario A (Product Page):

- Product image (with zoom/pan capability).
- Product name and brand.
- Price (including discount and offer badges).
- Detailed description and technical specifications.
- Customer reviews, ratings, and Q&A.
- Quantity selector and stock indicator.
- "Add to Cart" and "Buy Now" buttons.
- Delivery options and estimated date.
- Related/suggested products carousel.
- Return policy and warranty info.

Why:

- Users depend on clear images and complete details before making purchasing decisions.
- Fast access to "Add to Cart" or "Buy Now" streamlines shopping.
- Related products increase cross-selling and customer satisfaction.

Extend for Other Scenarios:

- To-Do List App: Task input field, add/cancel buttons, task list, filters, completion checkboxes, delete/edit controls, sorting (date, priority).
 - Social Media Profile: Profile picture uploader, name/bio editing, follower/following counts, tabs for posts/photos, follow/unfollow button, connection request controls.
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3. Sketch Initial Low-Fidelity Wireframes

Purpose:

- To rapidly explore multiple interface layouts without investing significant time in details or visuals.
- Low-fidelity wireframes typically use rectangles, circles, and simple text to block out interface sections.

Task:

- For each main screen, sketch 2-3 layouts, focusing on the arrangement of headers, navigation, key buttons, content areas, and lists.
- For an e-commerce app, you might have:
 - Home Screen: Search bar at top, “Today’s Deals” banner, navigation tabs below, product categories in grid.
 - Product Listing: Filters/sorting at the top, grid of products with images and price.
 - Product Details: Large product image, price and “Add to Cart” button nearby, tabs for description/reviews.
 - Cart: List of items with price, delete/reduce quantity options.
 - Checkout: Form for address, payment summary, selection of payment options.
 - Confirmation: Order summary, tracking option, thank you note.

Why:

- Exploring multiple options avoids early commitment to a suboptimal layout.
 - Visualizes user path without distractions of visuals/colors.
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4. Choose and Document Optimal Layouts

Task:

- Review each wireframe, choosing those that offer the clearest information hierarchy and easiest navigation.
- Document your choice and the rationale in the lab record.

Sample Documentation:

- “Layout 2 for Product Details chosen due to clearer separation of image and description area, and proximity of ‘Add to Cart’ button to price info.”
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5. Define and Map Out User Flow

Purpose:

- User flow diagrams help visualize the sequence of steps users take to accomplish key tasks, ensuring no dead-ends or confusing transitions.

Example User Flow for Scenario A:

- Home → Browse by Category → Product Listing → View Product Details → Select Quantity → Add to Cart → View Cart → Proceed to Checkout → Choose Payment → Confirm Order → View Confirmation Screen

Annotation:

- Mark screens where users might hesitate or make mistakes (e.g., unclear “Add to Cart” placement, confusing filter options).
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6. Develop Mid-Fidelity Wireframes Using a Tool

With the optimal layout chosen, create wireframes using a digital tool, such as Figma, Balsamiq, or Adobe XD.

Elements to Include:

- Headers, footers, navigation bars for structure and accessibility.
- Placeholders for images (rectangles with an “X” or representative icon).
- Blocks for text (labels, headings, descriptions, product info, and messages).
- Buttons for actions (“Add to Cart,” “Proceed to Checkout,” “New Task”).
- Input fields for search and forms (address entry, payment details).
- Lists/grids for displaying content (product grids, task lists, follower lists).
- Basic icons for navigation, actions, and status indicators.

Explanation:

- A mid-fidelity wireframe strips away unnecessary styling but includes real labels, UI elements, and interaction affordances needed for prototyping and testing.
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7. Layout, Spacing, and Information Hierarchy

Explanation:

- Arrange elements so users can easily find what they need and perform primary actions with minimal effort.
- Use larger or highlighted areas for crucial elements (e.g., the “Buy Now” button), group related information together (description next to price), and maintain enough whitespace for clarity.

Example:

- “The Product Details page places the image at the top, with price and cart options directly below, then reviews and related products in tabs to the side. Action buttons are large and colored for emphasis.”

8. Tabulate User Interaction and Usability Feedback

Field	Detailed Example Entry and Reasoning
Name of Product	Flipkart Mobile App
Reasons for hesitation	Users paused when filters were hidden; some were unsure about how to view detailed specs.
Reasons for mistakes	Selecting wrong color/variant due to unclear dropdown; accidental double-add to cart.
Source for confusion	Similar “Buy Now” and “Add to Cart” buttons side by side; vague address fields in checkout.
Time required to complete the task	2 days, more than expected-users spent extra time comparing prices in cart vs product pages.
Help and assistant	Help link was in footer (hard to find); some users didn’t know FAQs existed; wording of prompts was not precise.
Product worked as expected	Yes (for majority), but some experienced failed redirects after checkout.

9. Identify Common Usability Issues

Review all feedback and observations to find recurring problems.

- Navigation elements are not always visible or accessible from every screen.
- Button and link labels are ambiguous (“Submit” vs “Place Order”).
- Important information (stock levels, delivery date) not shown upfront.
- Help/support is hidden or unclear; users struggle to find assistance.
- Tasks (purchase, add, edit) sometimes require too many steps.
- Error messages are vague, lacking guidance for correction.
- Overwhelming interface: too many actions on one screen.
- Feedback after major actions is weak (e.g., order placed, task added).
- Inconsistent placement of critical buttons leads to misclicks.
- Mobile responsiveness issues-elements overlap or get cut off.

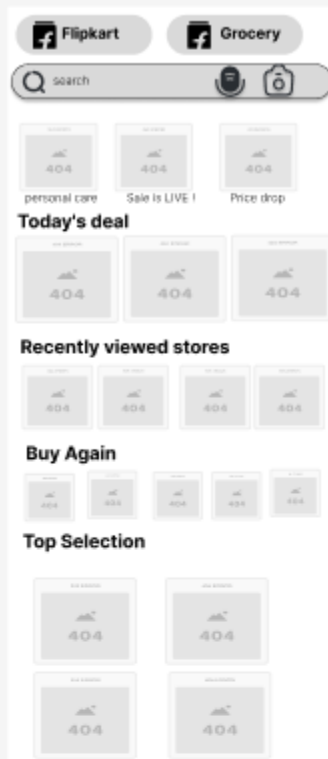
10. Prevention Strategies During Design & Engineering

Reflect and propose actionable steps that could have avoided the identified issues:

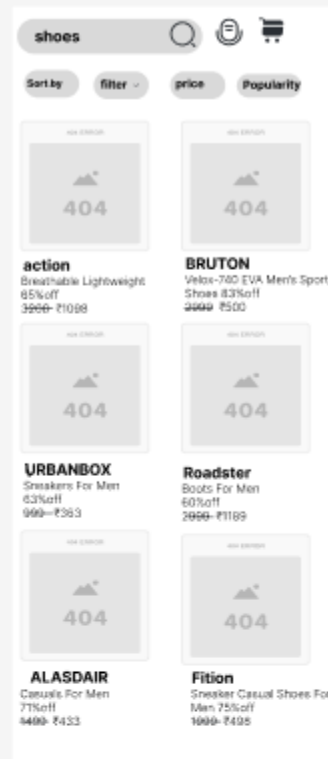
- Involve real users for early prototype testing: Gather feedback on wireframes before beginning final design or development.
- Consistent labeling and terminology: Use clear, universally understood words for buttons and links.
- Intuitive layouts: Place high-priority actions in prominent, predictable positions (e.g., bottom-right for “Buy Now”).
- Accessibility: Ensure help and support links are always accessible, and feedback messages are clear.
- Progressive disclosure: Limit information presented at once; use tabs, accordions, or overlays for optional details.
- Minimal steps: Use one-click actions where possible to streamline flows.
- Clear error handling: Provide specific, actionable error messages.
- Responsive design: Test wireframes on all device types for usability.
- Visual hierarchy: Use size, color, and spacing to indicate importance.
- Interactive prototyping: Validate user flows with clickable wireframes before building the full product.

Link: <https://www.flipkart.com/>

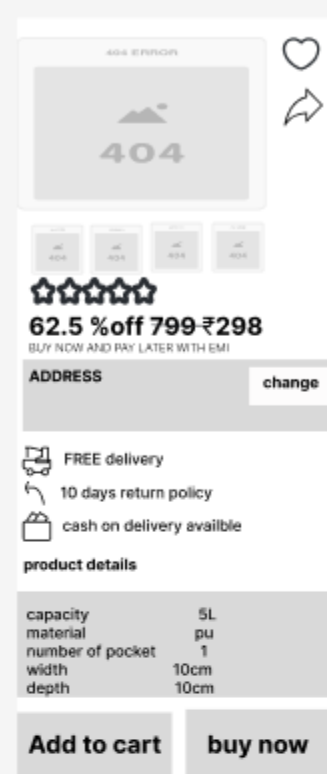
Home Screen



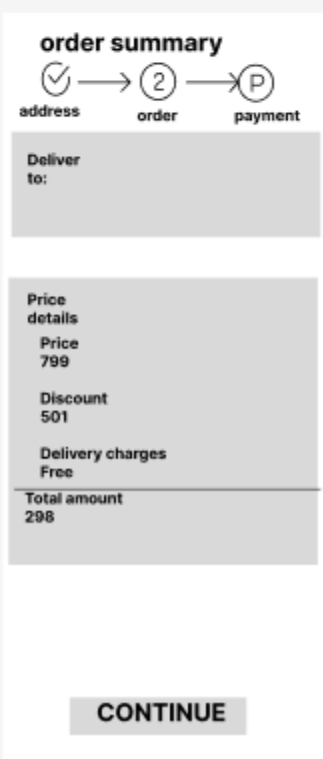
Product Listing



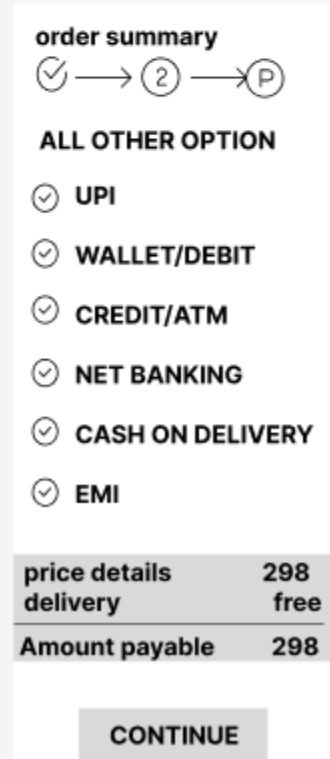
Product Details Page



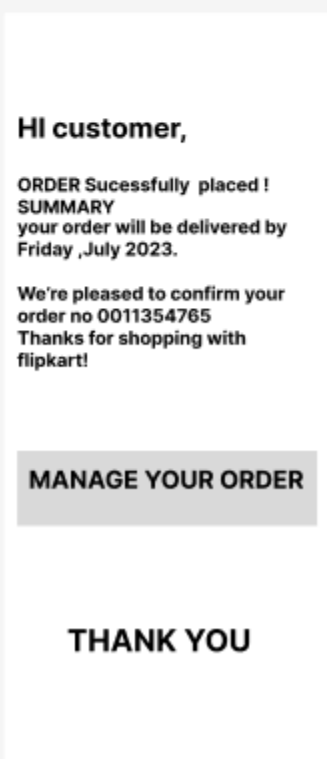
Checkout – Order Summary



Payment Options



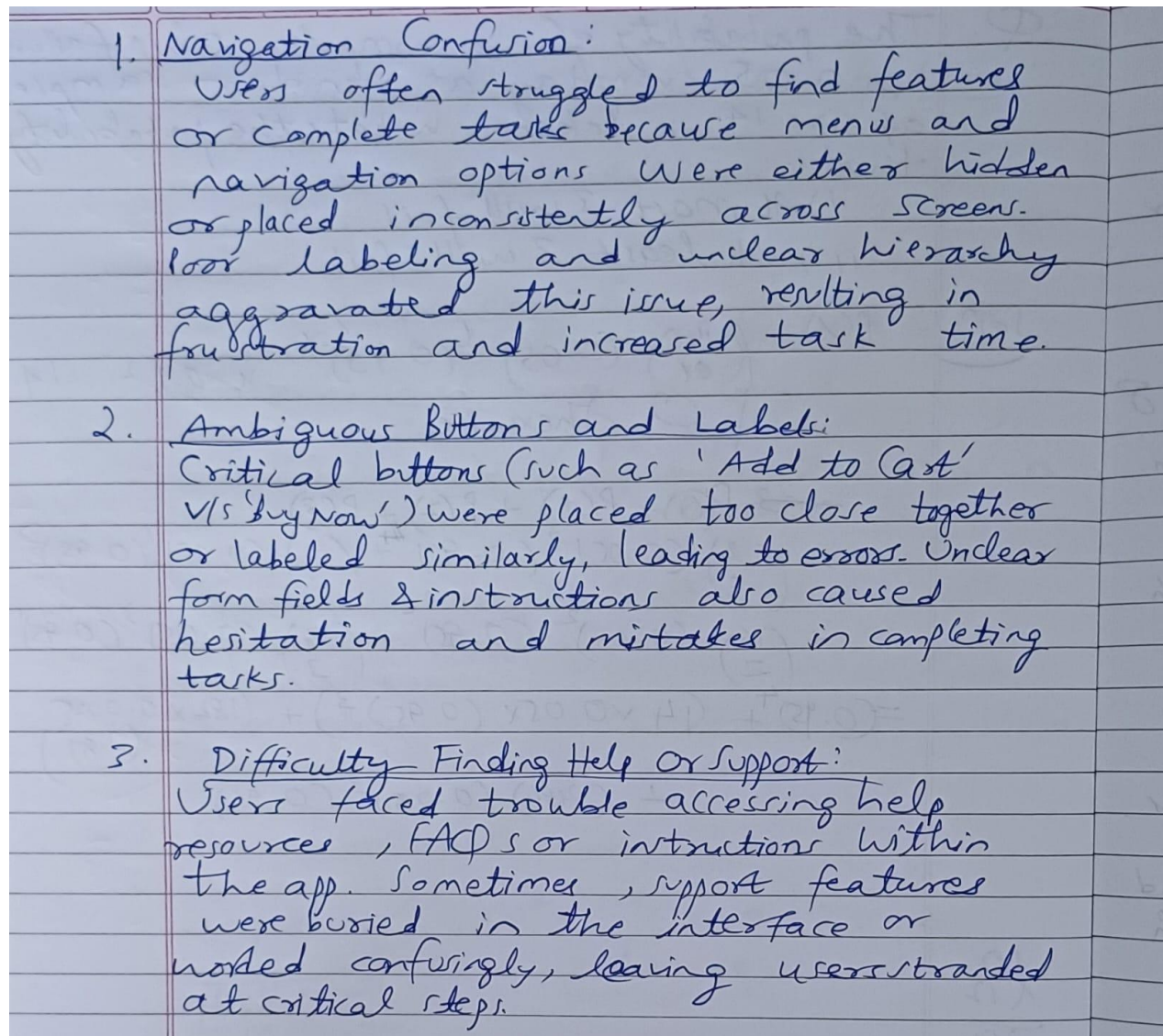
Order Confirmation



Post Laboratory Questions:

Following questions are to be handwritten and attach photograph of the answer in this document.

1. What were the top three most common types of interaction troubles encountered?



2. Impact of poor design in user experience.

Poor design in a product directly undermines the user's ability to achieve goals efficiently and comfortably. It leads to increased error rates, longer task completion times, and overall dissatisfaction. Users may abandon products that're confusing or frustrating, resulting in negative reviews and loss of engagement or

business. Additionally, poor design can exclude users with disabilities or those using mobile devices, decreasing accessibility and inclusivity.

3. Define User Experience (UX) in your own words, distinguishing it from User Interface (UI).

UX refers to the total perception and satisfaction that a person has while interacting with a product, system or service. It encompasses emotional responses, ease of use and the feeling of achieving intended goals without frustration.

Distinguishing UX & UI:

UX is about how a product works- how intuitive, responsive and enjoyable the journey feels. UI concerns how a product looks- the arrangement of buttons, colors, typography and visual styles. While UI is a subset of UX, UX looks at the broader picture- the value, utility and emotional impact over every interaction.

4. Name and briefly describe two key components or aspects that contribute to a positive user experience.

1. Ease of use (usability):- The product should be straightforward, intuitive and efficient, allowing users to accomplish tasks quickly with minimal learning curve & few mistakes.

2. Accessibility:- Everyone, including people with disabilities, should be able to use the product. This means providing alternative text for images, keyboard navigation, sufficient color contrast and designs that work across devices & screen sizes.

5. Why is understanding the user's context and emotions important when designing for user experience?

Understanding a user's circumstances, environment and emotional state helps designers create products that anticipate real-world needs and frustrations. For example, a hurried shopper values speed & clarity, while someone searching for medical advice requires reassurance & simplicity. Not considering emotions can lead to cold, unwelcoming interfaces that ignore genuine user struggles, decreasing trust & satisfaction.

6. How do usability and accessibility contribute to the overall user experience of a product?

Usability ensures that users can accomplish desired tasks efficiently, effectively and without error.

Accessibility guarantees that all users, regardless of ability or context, can engage with the product's core features. Both are essential for positive UX: they minimize frustration, foster inclusivity and expand the product's reach. Designing with these principles avoids excluding large segments of users and builds loyalty by making products genuinely helpful to everyone.